Physics 5450: Midterm Exam II

(Dated: November 27, 2007)

Partial credit is given according to the point count stated after each question.

1. Consider the *two-dimensional* harmonic oscillator, for which the potential energy is

$$V(x,y) = \frac{1}{2}m\omega^2(x^2+y^2)$$

(a) Show that separation of variables in cartesian coordinates turns this into two one-dimensional oscillators, and exploit the knowledge of the latter to determine the allowed energies E_n. [10 pts.]

(b) Determine the degeneracy $d(n)$ of the n-th level E_n. [10 pts.]

(c) Write explicit, normalized wave functions for cases $n=0$ and $n=1$. Re-express your results in terms of polar coordinates: radius $\rho = \sqrt{x^2+y^2}$ and polar angle φ. [10 pts.]

Bonus: this question requires careful thinking – return to it once you worked on all other problems:

(d) Using results from part (c) for $n=1$ level, find the effect of the angular momentum operator $L_z = xp_y - yp_x$ (it may be helpful to recall that in polar coordinates $L_z = -i\hbar\frac{\partial}{\partial\varphi}$) on these wave functions. Use these results to form linear superposition that are *eigenstates* of L_z. What are their respective eigenvalues? [10 pts.]

2. Electron is in the $n=2, \ell=m=0$ state of hydrogen, ψ_{200}. (See Table 4.7.)

(a) What is the *most probable* value of r? *Hint:* First you must figure out the probability that the electron is found between r and $r+dr$. [20 pts.]

(b) What is the *least probable* value of r, in addition to the two obvious, $r=0$ and $r=\infty$, least likely values? [10 pts.]

3. The hydrogen atom is in the *ground state.* Your task is to calculate the probability of finding the electron inside the classically forbidden region (classically forbidden is the region where the potential energy exceeds the total energy of an electron) by following these steps:

(a) Sketch the potential energy for this state, and indicate the location of the classical turning point r_0 on the graph. [5 pts.]

(b) Calculate r_0. Express your result for the classical turning point in terms of Bohr radius a. [5 pts.]

(c) Represent the probability in question in terms of dimensionless integral and evaluate it. [10 pts.]

4. Table 4.3 shows that $Y_2^1 = -\sqrt{15/8\pi}\sin\theta\cos\theta e^{+i\phi}$.

(a) Apply the raising operator L_+ (eq.4.130) to find $Y_2^2(\theta,\phi)$. Use equation 4.121 to get the normalization. [10 pts.]

(b) Apply the lowering operator L_- (eq.4.130) to find $Y_2^0(\theta,\phi)$. Use equation 4.121 to get the normalization. [10 pts.]

P1 $V(x,y) = \frac{1}{2} m\omega^2 (x^2+y^2)$

(a) $$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\psi(x,y) + \frac{1}{2}m\omega^2(x^2+y^2)\,\psi(x,y) = E\,\psi(x,y)$$

Separation of variables $\psi(x,y) = X(x)\,Y(y)$: plug in S.E. and divide by XY

$$\underbrace{\frac{1}{X}\left(-\frac{\hbar^2}{2m}\frac{d^2X}{dx^2}+\frac{1}{2}m\omega^2x^2X\right)}_{E_x} + \underbrace{\frac{1}{Y}\left(-\frac{\hbar^2}{2m}\frac{d^2Y}{dy^2}+\frac{1}{2}m\omega^2y^2Y\right)}_{E_y} = E$$

$$\Rightarrow \begin{cases} -\frac{\hbar^2}{2m}\frac{d^2X}{dx^2}+\frac{1}{2}m\omega^2x^2X = E_x X \\ -\frac{\hbar^2}{2m}\frac{d^2Y}{dy^2}+\frac{1}{2}m\omega^2y^2Y = E_y Y \end{cases} \qquad E_x + E_y = E$$

But $E_x = \hbar\omega\left(n_x+\frac{1}{2}\right)$, $E_y = \hbar\omega\left(n_y+\frac{1}{2}\right)$ as both describe one-dimensional harmonic oscillators

$n_x, n_y \in$ Integers

$$\Rightarrow E = E_x + E_y = \hbar\omega(\underbrace{n_x+n_y}_{n}+\tfrac{1}{2}+\tfrac{1}{2}) = \hbar\omega(n+1)$$

$n \in$ Integer

(b) $E_n = \hbar\omega(n+1)$, $n = n_x + n_y$

Degeneracy $d(n)$ is found by: for given fixed $\underline{n}$ we can take

$$(n+1)\text{ ways}\begin{cases} n_x = n & n_y = 0 \\ n_x = n-1 & n_y = 1 \\ n_x = n-2 & n_y = 2 \\ \vdots & \vdots \\ n_x = 1 & n_y = n-1 \\ n_x = 0 & n_y = n \end{cases}$$

$\Rightarrow d(n) = n+1$

(c) $n=0 \Rightarrow n_x = n_y = 0$: this is the ground state $E_0 = \hbar\omega$

$$\psi_0(x,y) = \psi_0(x)\,\psi_0(y) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}y^2}$$

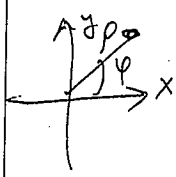
$$\Rightarrow \psi_0(x,y) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{-\frac{m\omega}{2\hbar}\rho^2} \quad , \rho^2 = x^2+y^2$$

$\underline{n=1}$ $\quad n_x = 1 \quad n_y = 0 \Rightarrow \psi_{10}$
$\quad n_x = 0 \quad n_y = 1 \Rightarrow \psi_{01}$ $\Big\}$ two-fold degenerate state

Using that first excited state of 1d oscillator is described by (eq. 2.62) $\psi_1 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \sqrt{\frac{2m\omega}{\hbar}}\, x\, e^{-\frac{m\omega}{2\hbar}x^2}$

we have

$$\psi_{10}(x,y) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\left(\frac{2m\omega}{\hbar}\right)^{1/2} x\, e^{-\frac{m\omega}{2\hbar}x^2}\, e^{-\frac{m\omega}{2\hbar}y^2} =$$



$$= \sqrt{\frac{2}{\pi}}\, \frac{m\omega}{\hbar}\, \rho\cos\varphi\, e^{-\frac{m\omega}{2\hbar}\rho^2} \quad , x = \rho\cos\varphi$$

$$\psi_{01}(x,y) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\left(\frac{2m\omega}{\hbar}\right)^{1/2} e^{-\frac{m\omega}{2\hbar}x^2}\, y\, e^{-\frac{m\omega}{2\hbar}y^2} =$$

$$= \sqrt{\frac{2}{\pi}}\, \frac{m\omega}{\hbar}\, \rho\sin\varphi\, e^{-\frac{m\omega}{2\hbar}\rho^2} \quad , y = \rho\sin\varphi$$

(d) $\hat{L}_z = x p_y - y p_x = -i\hbar \frac{\partial}{\partial\varphi}$

$$\Rightarrow \hat{L}_z \psi_{10} = \sqrt{\frac{2}{\pi}}\, \frac{m\omega}{\hbar}\, (i\hbar)\, \rho\sin\varphi\, e^{-\frac{m\omega}{2\hbar}\rho^2} = i\hbar\, \psi_{01}$$

$$\hat{L}_z \psi_{01} = \sqrt{\frac{2}{\pi}}\, \frac{m\omega}{\hbar}\, (-i\hbar)\, \rho\cos\varphi\, e^{-\frac{m\omega}{2\hbar}\rho^2} = -i\hbar\, \psi_{10}$$

This is because $\frac{\partial}{\partial\varphi}\cos\varphi = -\sin\varphi$ but $\frac{\partial}{\partial\varphi}\sin\varphi = \cos\varphi$.

$\Rightarrow$ form $\psi_\pm = \psi_{10} \pm i\psi_{01}$

Then $\hat{L}_z \psi_\pm = \hat{L}_z(\psi_{10} \pm i\psi_{01}) = i\hbar\psi_{01} \pm i(-i\hbar)\psi_{10} =$
$= \pm\hbar\psi_{10} + i\hbar\psi_{01} = \pm\hbar(\psi_{10} \pm i\psi_{01}) =$
$= \pm\hbar\psi_\pm$

$\Rightarrow \psi_+$ is eigenstate of $\hat{L}_z$ with eigenvalue $+\hbar$
ψ_- with $-\hbar$

In fact, $\psi_\pm = \sqrt{\frac{2}{\pi}}\, \frac{m\omega}{\hbar}\, \rho\, e^{\pm i\varphi}\, e^{-\frac{m\omega}{2\hbar}\rho^2}$

P2

$$\psi_{200} = R_{20}\, Y_0^0 = \frac{1}{\sqrt{4\pi}}\,\frac{1}{\sqrt{2}}\, a^{-3/2}\left(1-\frac{r}{2a}\right) e^{-r/2a}$$

(a) $$\text{Prob}(r) = \underbrace{|\psi_{200}|^2}_{\text{probability density}}\ \underbrace{4\pi r^2 dr}_{\text{spherically symmetric volume (shell)}} = \frac{1}{2a^3}\left(1-\frac{r}{2a}\right)^2 e^{-r/a}\, r^2 dr$$

$$\Rightarrow \text{Prob}(r) = p(r)\,dr \quad , \quad p(r) = \frac{1}{2a^3}\, r^2 \left(1-\frac{r}{2a}\right)^2 e^{-r/a}$$

$\frac{dp(r)}{dr} = 0$ gives extrema points: maxima and minima

$$\frac{dp(r)}{dr} = \frac{1}{2a^3}\left\{ 2r\left(1-\frac{r}{2a}\right)^2 e^{-r/a} + r^2 \cdot 2\left(1-\frac{r}{2a}\right)\left(-\frac{1}{2a}\right) e^{-r/a} \right.$$

$$\left. + r^2\left(1-\frac{r}{2a}\right)^2 e^{-r/a}\left(-\frac{1}{a}\right)\right\} =$$

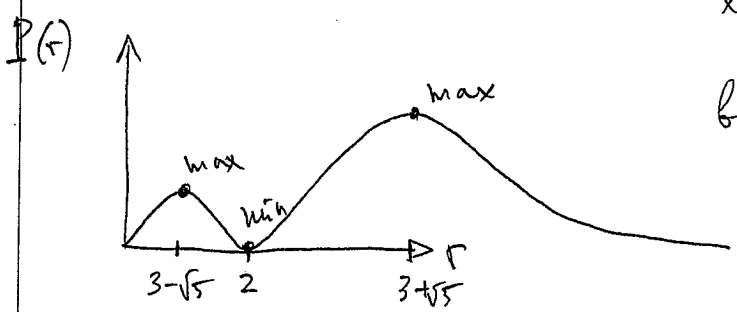
$$= \frac{1}{2a^3}\, r\left(1-\frac{r}{2a}\right) e^{-r/a}\left\{ 2\left(1-\frac{r}{2a}\right) - \frac{2r}{2a} - \frac{r}{a}\left(1-\frac{r}{2a}\right)\right\}$$

$$= 0$$

$$\Rightarrow \quad 2 - \frac{r}{a} - \frac{r}{a} - \frac{r}{a} + \frac{1}{2}\left(\frac{r}{a}\right)^2 = 0$$

$\Rightarrow$ solve for $x = \frac{r}{a}$: $\quad x^2 - 6x + 4 = 0$

$$x_{1,2} = 3 \pm \sqrt{9-4} = 3 \pm \sqrt{5}$$



both are points of maxima:

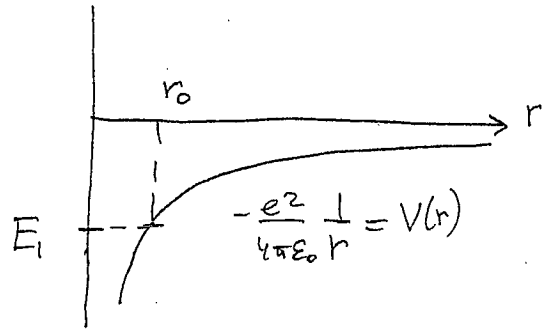
$r_1 = (3-\sqrt{5})\,a$; $r_2 = (3+\sqrt{5})\,a$

(Among them, r_2 actually gives bigger probability, but this was not part of the question)

(b) Observe that $p(r) = 0$ for $r = 2a$

$\Rightarrow r_3 = 2a$ is the least likely value.

P3



Ground state $n=1, l=0, m=0$

Since $l=0$, centrifugal potential $V_l = \frac{\hbar^2 l(l+1)}{2mr^2}$ is absent!

(a) Find r_0 from condition $E = \frac{p^2}{2m} + V = -\frac{e^2}{4\pi\varepsilon_0 r_0} = E_1$

$(p=0)$

Then $r_0 = -\frac{e^2}{4\pi\varepsilon_0 E_1}$, $E_1 = -\frac{m}{2\hbar^2}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2$

(b) $\Rightarrow r_0 = \frac{4\pi\varepsilon_0 \cdot 2\hbar^2}{e^2 \cdot m} = 2a$ where $a = \frac{4\pi\varepsilon_0\hbar^2}{me^2}$ Bohr radius

(c) Probability to be outside r_0 $\qquad \psi_{100} = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}$

$$P = \int_{r_0}^{\infty} |\psi_{100}|^2 r^2 dr \underbrace{\int_0^{\pi} \sin\theta d\theta \int_0^{2\pi} d\varphi}_{4\pi} =$$

$$= 4\pi \int_{r_0}^{\infty} \left(\frac{1}{\sqrt{\pi a^3}}\right)^2 \left(e^{-r/a}\right)^2 r^2 dr =$$

$$= \frac{4}{a^3} \int_{r_0=2a}^{\infty} r^2 e^{-2r/a} dr = \frac{4}{a^3} \int_4^{\infty} \left(\frac{a}{2}\right)^3 x^2 e^{-x} dx =$$

$r = \frac{a}{2}x$

$dr = \frac{a}{2}dx$

$$= \frac{1}{2} \int_4^{\infty} x^2 e^{-x} dx = \frac{1}{2} \int x^2 d(-e^{-x}) =$$

$$= \frac{1}{2}\left\{-x^2 e^{-x}\Big|_4^{\infty} + \int_4^{\infty} e^{-x} 2x dx\right\} = \frac{1}{2}\left\{-x^2 e^{-x}\Big|_4^{\infty}\right.$$

$$\left. - 2xe^{-x}\Big|_4^{\infty} + 2\int_4^{\infty} e^{-x} dx\right\} = \frac{1}{2}\left(4^2 e^{-4} + 2\cdot 4 e^{-4} + 2e^{-4}\right) =$$

$= 13e^{-4} \approx 0.24$ (not a small probability!)

P4 $\quad Y_2^1(\theta,\varphi) = -\sqrt{\frac{15}{8\pi}}\,\sin\theta\cos\theta\, e^{i\varphi}$

(a) from eq. 4.121 $\quad \hat{L}_+ Y_2^1 = (A_2^1)\, Y_2^2$

where $A_2^1 = \hbar\sqrt{2(2+1) - 1(1+1)} = 2\hbar$

$$\Rightarrow Y_2^2 = \frac{1}{2\hbar}\,\hbar\, e^{i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right) Y_2^1 =$$

$$= \frac{\hbar}{2\hbar}\left(-\sqrt{\frac{15}{8\pi}}\right) e^{i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right)\sin\theta\cos\theta\, e^{i\varphi} =$$

$$= \frac{\hbar}{2\hbar}\left(-\sqrt{\frac{15}{8\pi}}\right) e^{i\varphi\cdot 2}\left(\cos^2\theta - \sin^2\theta + \underbrace{i\frac{\cos\theta}{\sin\theta}\cdot i\cdot\sin\theta\cos\theta}_{-\cos^2\theta}\right) =$$

$$= \frac{\hbar}{2\hbar}\sqrt{\frac{15}{8\pi}}\,\sin^2\theta\, e^{i2\varphi} = \sqrt{\frac{15}{32\pi}}\,\sin^2\theta\, e^{i2\varphi}.$$

(b) from eq. 4.121 $\quad L_- Y_2^1 = (A_2^1)\, Y_2^0$

where $A_2^1 = \hbar\sqrt{2(2+1) - 1(1-1)} = \sqrt{6}\,\hbar$

$$\Rightarrow Y_2^0 = \frac{1}{\sqrt{6}\hbar}\,\hat{L}_- Y_2^1 = \frac{1}{\sqrt{6}\hbar}(-\hbar)\, e^{-i\varphi}\left(\frac{\partial}{\partial\theta} - i\cot\theta\frac{\partial}{\partial\varphi}\right)\cdot$$

$$\cdot\left(-\sqrt{\frac{15}{8\pi}}\right)\sin\theta\cos\theta\, e^{i\varphi} = \frac{1}{\sqrt{6}}\sqrt{\frac{15}{8\pi}}\, e^{-i\varphi} e^{i\varphi}\Big(\cos^2\theta - \sin^2\theta$$

$$\underbrace{- i\cot\theta\cdot i\cos\theta\sin\theta}_{+\cos^2\theta}\Big) = \sqrt{\frac{15}{48\pi}}\left(2\cos^2\theta - \sin^2\theta\right)$$

$$\Rightarrow Y_2^0 = \sqrt{\frac{5}{16\pi}}\left(3\cos^2\theta - 1\right).$$